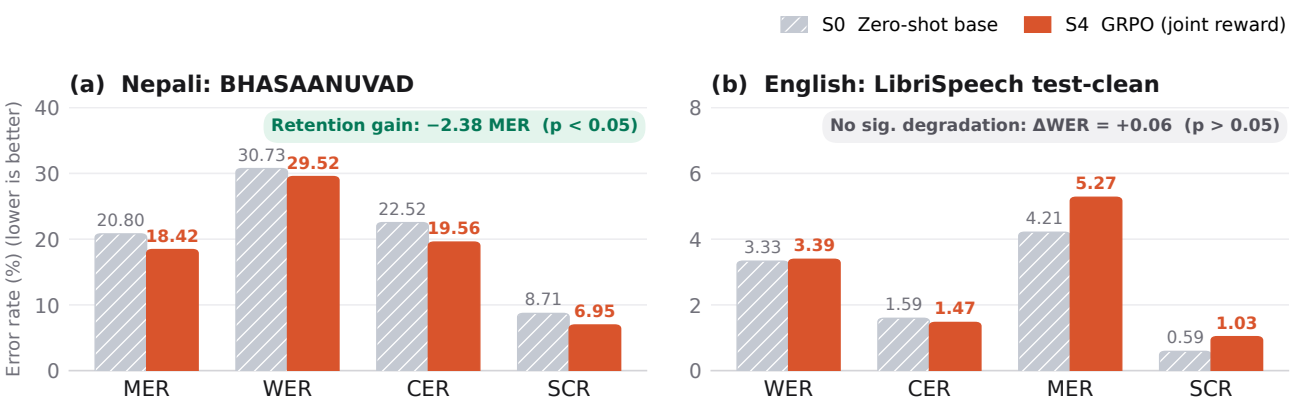
S0 Zero-shot base
S4 GRPO (joint reward)
(a) Nepali: BHASAANUVAD
Retention gain: −2.38 MER (p < 0.05)
Error rate (%) (lower is better)
40
30
20
10
0
20.80
18.42
30.73
29.52
22.52
19.56
8.71
6.95
MER
WER
CER
SCR
(b) English: LibriSpeech test-clean
No sig. degradation: ΔWER = +0.06 (p > 0.05)
8
6
4
2
0
3.33
3.39
1.59
1.47
4.21
5.27
0.59
1.03
WER
CER
MER
SCR